

BINF 4002

Lecture 1

Today's Class

1. Introductions!
2. Course Logistics
3. Goals
4. ~ Syllabus / Assignments / Exams
5. Activity

Course Logistics

1. Lectures Tue/Thu, 9-10:15
2. Recitations Mondays 9-10am (starting next week)
3. There will be 3 assignments, a Paper / Paper Discussion, and a Final Exam
4. Platforms:
 - a. Courseworks
 - b. Ed / Slack / canvas / harmonize / piazza (poll?)

CATEGORIES

- Indiv. Assignment 1
- Indiv. Assignment 2
- Indiv. Assignment 3
- Semester Project (Misc)
- Checkpoint 1
- Checkpoint 2
- Checkpoint 3
- Checkpoint 4
- Checkpoint 5
- Logistics
- Social
- Miscellaneous

Search

Filter

Dec 21, 2025

Demo Day

Indiv. Assignment 1 Anonymous 4w

1

Final Group Deliverable: Is it due today 12p...

Indiv. Assignment 1 Anonymous 4w

1

Not able to submit FINAL PROGRESS REPOR...

Indiv. Assignment 1 Jeffrey Roh 4w

1

Checkpoint 5 (Final Individual Project Grade)

Checkpoint 5 Anonymous 1mth

1

video for 16:9

Indiv. Assignment 1 Anonymous 1mth

1

Demo Video Question

Semester Project (Misc) Anonymous 1mth

1

IA1 Different rendering depending on browser #36

F

Florent Pollet

4 months ago in Indiv. Assignment 1

★

👁

152

STAR

WATCHING

VIEWS



Hi,

1

I realized the dropdown looks different between Safari (left) and Chrome (right). Is it good as is or should I try to fix the rendering in Safari (center it vertically or expand it)?

Thank you!

...

2 Answers

Brian A. Smith **STAFF**

4 months ago



This is fine. No need to worry about browsers using slightly different styling on form widgets.



Thanks for making sure, though!

...

A

Ali Zia

4 months ago



Same issue here.

1

...

Course Logistics

1. Lectures Tue/Thu, 9-10:15
2. Recitations Mondays 9-10am (starting next week)
3. There will be 3 assignments, a Paper / Paper Discussion, and a Final Exam
4. Platforms:
 - a. Courseworks
 - b. Ed / Slack / canvas / harmonize / piazza (poll?)
5. Question: Flipped classroom?

Goals & Expectations

It is *not* my goal with this course to give you the most direct path to practical readiness.

Goals & Expectations

It is *not* my goal with this course to give you the most direct path to practical readiness.

This is because that path changes rapidly as technology changes, and intuition/understanding will serve you better in the long term.

Goals & Expectations

My goal is to give you sufficient mathematical and computational understanding to have an effective intuition and a *health epistemic uncertainty* about topics in ML/AI.

Goals & Expectations

My goal is to give you sufficient mathematical and computational understanding to have an effective intuition and a *health epistemic uncertainty* about topics in ML/AI.

And secondarily, to have you appreciate why ML/AI for health/biomedicine is different than other areas of ML/AI.

Goals & Expectations

It is my expectation that you will learn independently, outside of class.

Goals & Expectations

It is my expectation that you will learn independently, outside of class.

This is because (a) we have too much background material to cover and (b) ChatGPT & other tools have lowered the barrier to independent learning.

Goals & Expectations

I will be presenting my views on this field, not necessarily the field consensus.

Goals & Expectations

I will be presenting my views on this field, not necessarily the field consensus.

This is because I'm teaching the course so I get to.

Goals & Expectations

It is my intent to be organized and proactive with this course.

Goals & Expectations

It is my *intent* to be organized and proactive with this course.

Goals & Expectations

I will treat all of you as though you are here to learn in good faith, and ask that you assume that I and the teaching staff are here to teach in good faith.

Goals & Expectations

Columbia policies trump anything I or others in the teaching staff say at any time.

Approximate Syllabus

1. Through February: Math background refresh.
2. Through spring break: ML Fundamentals refresh.
3. After spring break: ML for health specifically.

Approximate Syllabus

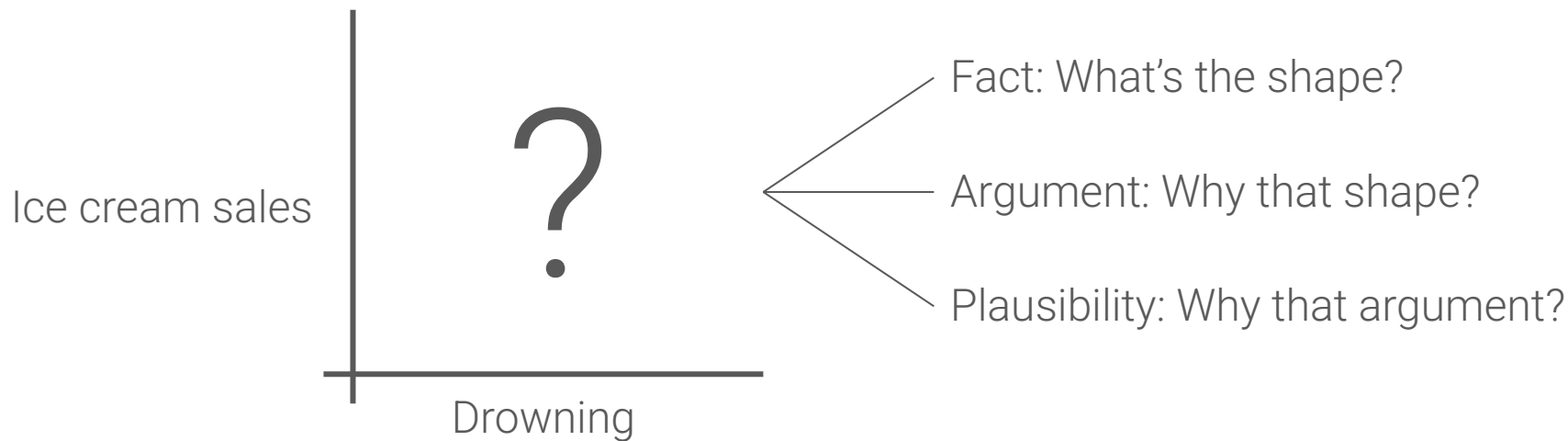
1. Through February: Math background refresh.
 - a. Linear algebra, probability, calculus
2. Through spring break: ML Fundamentals refresh.
 - a. Optimization, ML evaluation, training, development, data types, losses, models, and algorithms
3. After spring break: ML for health specifically.
 - a. Health data differences, health model differences, health deployments.

Assignments & exams

1. Likely 3 assignments, colab notebooks including coding and mathematical (latex) components.
2. A team paper on a deep dive of an ML topic (e.g., model type) and its current usage, followed by a discussion / oral exam on that paper.
3. A final exam.

Activity: Epistemic Humility

a.k.a. be confidently uncertain, be happy to be wrong



Activity! Initial assessment

1. 11 concepts / questions, 90 seconds per concept, one page per concept.
 - a. “Concept” will actually be set of related concepts
2. For each concept / question, attempt to (NOT IN ORDER):
 - a. Define the concept (formally, mathematically, operationally)
 - b. Describe the concept (qualitatively, with examples) and its key properties / related concepts.
 - c. Draw the concept (with a simple cartoon)
3. Not graded, not expecting you to finish everything. The goal of this is to calibrate the teaching staff on what you know.

1. Vector / Vector Space /
Linear Operator / Matrix?

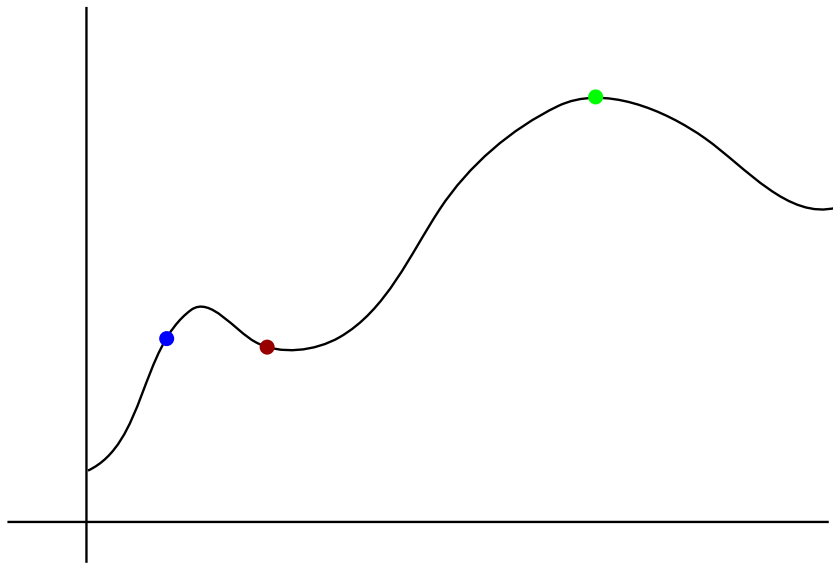
2. Random variable / PDF /
PMF / Expectation?

3. Joint / Conditional /
Marginal Random
variable?

4. Information content, Entropy, Mutual information?

5. Gradient?

Also draw the gradient on this function at the indicated points



6. Stationary point,
inflection point, local vs.
global minima vs
maxima, saddle point?

7. Gradient descent /
stochastic gradient
descent?

8. Latent variable,
expectation
maximization?

9. Binary classification,
confusion matrix,
AUROC?

10. k-nn, logistic regression,
decision tree, ensemble
method, neural network

11. Data types: Tabular data, regular grid data, ordinal sequences, time series, temporal point process/event stream.

Also highlight models useful for these data types.